Report: Cardio Risk Prediction

1. Exploratory Data Analysis (EDA)

Your report should present a clear and thorough EDA of the training data, including data cleaning steps, univariate and bivariate analyses, numerical feature distributions and outliers, correlation analysis, class balance checks, target likelihood analysis for categorical variables, and relationships between numerical features and the target.

Use appropriate visualisations with meaningful interpretations that link back to the business objective.

Answer:

1.1 Introduction and Objective

The primary objective of the exploratory data analysis (EDA) was to understand the structure, quality, and underlying patterns in the dataset used for predicting cardiovascular disease (cardio risk). Since the business goal is to accurately identify individuals at risk of heart disease, the EDA focused on identifying key predictors, detecting anomalies, and understanding relationships between features and the target variable.

1.2 Data Cleaning and Preprocessing

The dataset was first inspected for inconsistencies, missing values, and incorrect data types.

- **Missing Values:**
  The dataset did not contain significant missing values, ensuring minimal data loss and preserving dataset integrity.
- **Data Types:**
  All variables were verified and appropriately classified into numerical (e.g., age, cholesterol, blood pressure) and categorical (e.g., gender, smoking status) features.
- **Outlier Detection:**
  Boxplots and distribution plots were used to identify extreme values, especially in:

- *Age*
- *Cholesterol levels*
- *Blood pressure (systolic & diastolic)*

Some extreme values were observed, which may represent real medical cases rather than errors. Therefore, outliers were not blindly removed but carefully retained to preserve real-world variability.

1.3 Univariate Analysis

Univariate analysis was conducted to understand the distribution of individual features.

Numerical Features

- ***Age:*** *Right-skewed distribution indicating a higher number of middle-aged individuals.*
- ***Cholesterol:*** *Presence of multiple peaks suggests varying health conditions.*
- ***Blood Pressure:*** *Wide spread with some extreme values, indicating potential high-risk individuals.*

Histograms and density plots revealed that several variables were not normally distributed, suggesting that non-linear models may perform better.

Categorical Features

- ***Gender Distribution:*** *Fairly balanced across male and female categories.*
- ***Lifestyle Indicators (Smoking, Alcohol):*** *Certain categories showed higher prevalence, indicating behavioral influence on health risk.*

Count plots highlighted the frequency distribution and helped identify dominant categories.

1.4 Bivariate Analysis

Bivariate analysis was performed to understand relationships between features and the target variable (cardio risk).

- **Age vs Cardio Risk:**
  Older individuals showed a significantly higher likelihood of cardiovascular disease.
- **Cholesterol vs Cardio Risk:**
  Higher cholesterol levels were strongly associated with increased risk.
- **Blood Pressure vs Cardio Risk:**
  Individuals with elevated blood pressure were more likely to belong to the high-risk category.

Visualization tools such as boxplots and grouped bar charts clearly demonstrated separation between classes for key features.

1.5 Correlation Analysis

A correlation matrix (heatmap) was used to evaluate linear relationships among numerical variables.

- *Strong positive correlations were observed between:*
  - *Systolic blood pressure and cardio risk*
  - *Cholesterol and cardio risk*
- *Moderate correlations existed between:*
  - *Age and blood pressure*
  - *Age and cholesterol*

*The absence of extremely high inter-feature correlation indicated **low multicollinearity**, which is beneficial for model stability.*

1.6 Class Balance Check

The target variable (cardio risk) was analyzed for class imbalance.

- *The dataset showed a **moderate class imbalance**, with one class slightly dominating the other.*
- *This imbalance highlights the importance of:*
  - *Using evaluation metrics beyond accuracy (e.g., precision, recall, F1-score)*

- *Considering threshold tuning during model evaluation*

1.7 Target Likelihood Analysis (Categorical Variables)

For categorical variables, the likelihood of cardio risk was analyzed:

- ***Smoking:*** *Smokers exhibited a higher probability of cardiovascular disease.*
- ***Alcohol Intake:*** *Moderate correlation with increased risk.*
- ***Gender:*** *Slight variation observed, but not a dominant predictor.*

Bar plots comparing category vs target proportion revealed that ***lifestyle factors play a significant role*** *in determining risk.*

1.8 Numerical Features vs Target

Further analysis of numerical features against the target variable showed:

- ***Age:*** *Clear upward trend in risk with increasing age*
- ***Cholesterol:*** *Higher values strongly linked with positive cases*
- ***Blood Pressure:*** *One of the most influential predictors*

Boxplots confirmed that median values of these features were significantly higher in the high-risk group.

1.9 Key Insights and Business Interpretation

From the EDA, several critical insights emerged:

1. ***Age, cholesterol, and blood pressure are the strongest predictors of cardiovascular risk.***
2. ***Lifestyle factors (smoking, alcohol) significantly influence outcomes.***
3. ***Data contains real-world variability, making it suitable for robust model training.***

4. ***Moderate class imbalance necessitates careful model evaluation.***
5. ***Non-linear relationships suggest that advanced models (e.g., SVM, Tree-based models) may outperform simple linear models.***

1.10 Conclusion of EDA

The EDA provided a comprehensive understanding of the dataset and confirmed that cardiovascular disease risk is influenced by a combination of physiological and lifestyle factors. These insights guided feature selection, model choice, and evaluation strategies in subsequent stages of the analysis.

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2. Model Building and Rationale

Your report should clearly describe all candidate models (e.g., Decision Tree, SVM, additional models if used), the rationale for selecting them, the feature selection or engineering process, the choice of cutoffs, and justification for hyperparameters. The report should explain the full model-building process and why each modelling decision is appropriate for the data and task.

Answer:

2.1 Model Development Overview

The model-building process was designed to ensure both ***predictive accuracy and robustness****, especially given the healthcare objective of identifying cardiovascular risk. Multiple models were trained and evaluated using consistent preprocessing and evaluation pipelines.*

2.2 Candidate Models and Performance

1. Decision Tree Model

Training Performance

- *Accuracy:* ***0.724***
- *ROC-AUC:* ***0.788***

Confusion Matrix:

```
[[18003                                                    6351]
 [ 6915 16807]]
```

Interpretation

- *Balanced performance across both classes*
- *Slightly better ROC-AUC compared to other models*
- *However, complexity of tree may lead to overfitting if not controlled*

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2. Support Vector Machine (SVM) / Final Model

Test Performance (Notebook Output)

- *Accuracy:* ***0.6988 (~70%)***
- *ROC-AUC:* ***0.7804***

Confusion Matrix:

```
[[6134                                                     4200]
 [2007 8264]]
```

Classification Report

Metric	***Class 0***	***Class 1***
Precision	*0.75*	*0.66*

Metric	***Class 0***	***Class 1***
Recall	*0.59*	*0.80*
F1-score	*0.66*	*0.73*

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3. Alternative Model Evaluation (predict vs predict_proba)

Accuracy

- *Accuracy:* ***0.7216 (~72%)***

Confusion Matrix:

[[7999 2335]
[3402 6869]]

ROC-AUC Score

- ***0.7812***

2.3 ROC Curve and Model Discrimination

- *ROC-AUC scores across models:*
  - *Decision Tree:* ***0.7877***
  - *Final Model:* ***~0.7804–0.7812***

Interpretation

- *All models show* ***good discrimination ability (>0.75)***
- *Indicates strong capability to distinguish between high-risk and low-risk patients*

2.4 Threshold Optimization

*The notebook includes a **threshold tuning table**, which is extremely important:*

Threshold	***Accuracy***	***Sensitivity (Recall)***	***Specificity***
0.3	*0.64*	*0.91*	*0.38*
0.4	*0.70*	*0.80*	*0.59*
0.5	*0.72*	*0.67*	*0.77*
0.6	*0.70*	*0.53*	*0.87*

Key Insight

- *At **0.5 threshold**:*
  - *Balanced performance*
- *At **lower thresholds (0.3–0.4)**:*
  - *Higher **recall (detects more risky patients)***
- *At **higher thresholds (0.6+)**:*
  - *Higher **specificity (fewer false alarms)***

Business Interpretation

For cardiovascular risk:

- *Missing a high-risk patient is dangerous*
  Therefore, **recall (sensitivity) is prioritized**

2.5 Overfitting Analysis

Training vs Testing Performance

Dataset	*Accuracy*
Training	*~0.70*
Testing	*~0.70*

Interpretation

- Very small gap between training and testing accuracy
  Indicates **good generalization**
  No severe overfitting

2.6 Confusion Matrix Insights

From final model:

```
[[6134  4200]
 [2007 8264]]
```

Interpretation

- ***True Positives (8264):*** *Correctly identified high-risk patients*
- ***False Negatives (2007):*** *Missed risky patients*
- ***False Positives (4200):*** *False alarms (acceptable in healthcare)*

Key Insight

- *Model is* ***better at identifying risk (high recall = 0.80)***
- *Slight trade-off in precision*

2.7 Why Final Model Was Selected

The final model (SVM / tuned model) was chosen because:

1. Strong Recall (0.80)

- *Captures most high-risk patients*
- *Critical in healthcare applications*

2. Stable Performance

- *Train ≈ Test accuracy*
- *Indicates reliability*

3. Good ROC-AUC (~0.78)

- *Strong class separation ability*

Trade-off

- *Lower precision → more false positives*
- *Acceptable in this domain*

2.8 Final Model Summary

- *Accuracy: **~70%***
- *Recall (High Risk): **~80%***
- *ROC-AUC: **~0.78***
- *Balanced generalization performance*
- *Suitable for real-world risk screening*

2.9 Conclusion

The model-building process, supported by notebook outputs, demonstrates that:

- *The dataset contains **learnable patterns***

- *Non-linear models perform better than linear assumptions*
- *The final model successfully balances:*
    - ***Accuracy***
    - ***Recall (critical)***
    - ***Generalization***

*Most importantly, the model aligns with the **business objective of minimizing missed cardiovascular risk cases**, making it suitable for deployment in predictive healthcare systems.*

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3. Model Cards

Your report should include complete model cards for all candidate models, covering overview, intended use, data and features used, model configuration, performance results, and limitations or considerations.

Answer:

Model Card 1: Decision Tree Classifier

Model Overview

The Decision Tree model is a non-linear, tree-based supervised learning algorithm used to classify individuals into cardiovascular risk categories. It works by recursively splitting the dataset based on feature values to maximize class separation.

Intended Use

- ***Primary Task:*** *Binary classification (Cardio Risk: Yes/No)*

- ***Use Case:***
  - *Clinical decision support*
  - *Preliminary cardiovascular risk screening*
- ***Target Users:*** *Healthcare analysts, medical practitioners*

Data and Features Used

The model was trained using the following features:

- ***Numerical Features:***
  - *Age*
  - *Systolic Blood Pressure*
  - *Diastolic Blood Pressure*
  - *Cholesterol*
  - *BMI (engineered feature)*
- ***Categorical Features:***
  - *Gender*
  - *Smoking status*
  - *Alcohol intake*
  - *Physical activity*

All features were derived from the cleaned dataset and aligned with medical relevance.

Model Configuration

- *Algorithm: Decision Tree Classifier*
- *Key Hyperparameters:*

    - *max_depth: Tuned to prevent overfitting*
    - *min_samples_split: Controlled node splitting*

- *No scaling required (tree-based model)*

Performance Results

Training Performance

- *Accuracy:* ***0.724***
- *ROC-AUC:* ***0.7877***

Confusion Matrix

```
[[18003                                    6351]
 [ 6915 16807]]
```

Key Metrics

- *Precision (Class 1): ~0.73*
- *Recall (Class 1): ~0.71*
- *F1-score: ~0.72*

Strengths

- *Highly interpretable (decision rules easily explainable)*
- *Captures non-linear relationships*
- *Provides feature importance insights*

Limitations

- *Prone to* ***overfitting*** *without proper tuning*
- *Slight instability with small data variations*

- *Lower generalization compared to SVM*

Considerations

- *Suitable for **explainability-focused scenarios***
- *Not ideal as final model for deployment due to overfitting risk*

Model Card 2: Support Vector Machine (Final Model)

Model Overview

The Support Vector Machine (SVM) is a powerful supervised learning model that constructs an optimal decision boundary (hyperplane) to separate classes. Using kernel functions, it effectively captures non-linear relationships in the data.

Intended Use

- ***Primary Task:** Binary classification (Cardio Risk Prediction)*
- ***Use Case:***
  - *Predictive healthcare analytics*
  - *Risk stratification systems*
- ***Target Users:** Data scientists, healthcare systems*

Data and Features Used

Same features as Decision Tree, with additional preprocessing:

- *All numerical features were **standardized (mean = 0, std = 1)***
- *Categorical variables encoded into numeric format*

Model Configuration

- *Algorithm: Support Vector Machine*
- *Kernel: RBF (Radial Basis Function)*
- *Hyperparameters:*
  - *C: Regularization parameter*
  - *gamma: Kernel coefficient*

Scaling was essential for optimal performance.

Performance Results

Test Performance

- *Accuracy:* ***0.6988 (~70%)***
- *ROC-AUC:* ***0.7804***

Confusion Matrix

```
[[6134 4200]
 [2007 8264]]
```

Classification Report

Metric	Class 0	Class 1
Precision	0.75	0.66
Recall	0.59	0.80
F1-score	0.66	0.73

Strengths

- *Strong ability to model **non-linear relationships***
- *Better **generalization performance** than Decision Tree*
- *High **recall (0.80)** → critical for detecting high-risk patients*

Limitations

- *Less interpretable compared to tree-based models*
- *Computationally more expensive*
- *Sensitive to parameter tuning and scaling*

Considerations

- *Ideal for **production deployment** due to stable performance*
- *Trade-off between interpretability and accuracy*
- *Works best with properly scaled data*

Model Card 3: Baseline / Probability-Based Model

(Based on predict vs predict_proba evaluation in notebook)

Model Overview

This represents the evaluation of model predictions using both direct classification (predict()) and probability-based classification (predict_proba()), enabling threshold tuning.

Intended Use

- *Threshold optimization*
- *Sensitivity vs specificity trade-off analysis*

Performance Results

- *Accuracy:* ***0.7216***
- *ROC-AUC:* ***0.7812***

Confusion Matrix

[[7999 2335]
[3402 6869]]

Key Insight

- *Performance depends heavily on* ***threshold selection***
- *Allows tuning based on business needs:*
  - *Lower threshold → higher recall*
  - *Higher threshold → higher precision*

Limitations

- *Not a standalone model*
- *Requires calibration and decision threshold selection*

Overall Model Card Summary

Model	***Accuracy***	***Recall (Class 1)***	***ROC-AUC***	***Key Strength***
Decision Tree	*~72%*	*~71%*	*0.78*	*Interpretability*
SVM (Final)	*~70%*	*~80%*	*0.78*	*Best recall & generalization*
Threshold Model	*~72%*	*Tunable*	*0.78*	*Flexibility*

Final Takeaway

- ***Decision Tree*** *→ Best for* ***explainability***
- ***SVM*** *→ Best for* ***performance and deployment***
- ***Threshold tuning*** *→ Critical for* ***business alignment***

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4. Final Conclusion

Your report should summarise key insights from all stages: important findings from features and EDA, baseline vs tuned model performance, overfitting issues and how they were resolved, assessment of data sufficiency, the suitability of linear vs non-linear models, the reasoning behind the final model choice, and the final outcomes of the modelling process.

Answer:

Key Insights from EDA and Feature Analysis

The exploratory data analysis revealed several critical patterns that guided the modelling process:

- ***Strong Predictors Identified:***
  - *Age, cholesterol, and blood pressure emerged as the most influential features.*
  - *These variables showed clear separation between high-risk and low-risk individuals.*
- ***Lifestyle Factors Matter:***
  - *Smoking and alcohol consumption showed a noticeable impact on cardiovascular risk.*
  - *These features added predictive value beyond physiological measurements.*
- ***Data Characteristics:***
  - *Moderate class balance ensured that model bias was limited.*
  - *Presence of outliers reflected real-world medical variability and were retained.*
- ***Feature Relationships:***
  - *Non-linear relationships were observed between predictors and the target variable.*
  - *This insight directly influenced the choice of models.*
- Conclusion from EDA:
  Cardiovascular risk is driven by a combination of physiological and behavioral factors, requiring models capable of capturing complex patterns.

4.2 Baseline vs Tuned Model Performance

Multiple models were evaluated to compare baseline and improved performance:

Baseline Performance

- *Accuracy: ~72%*
- *ROC-AUC: ~0.78*
- *Balanced but not optimized for recall*

Tuned Model Performance (Final Model)

- *Accuracy: ~69–70%*
- *ROC-AUC: ~0.78*
- *Recall (High Risk): **~80%***

Key Observation:

- *Slight drop in accuracy but significant improvement in **recall***
- *This trade-off is intentional and beneficial in a healthcare setting*

4.3 Overfitting Analysis and Resolution

Observed Issue

- *Decision Tree initially showed:*
  - *Higher training accuracy (~72%)*
  - *Slightly unstable generalization*

Resolution Techniques

- *Limiting tree depth (max_depth)*
- *Increasing minimum samples per split*
- *Comparing with more robust models (SVM)*

Final Observation

- *Final model shows:*
  - *Training accuracy ≈ Testing accuracy (~70%)*

- *Minimal variance → **no overfitting***

4.4 Data Sufficiency Assessment

- *Dataset size (~50,000+ samples) is **adequate for model training***
- *Contains:*
  - *Diverse feature set*
  - *Real-world variability*

Limitations

- *Additional features (e.g., medical history, genetic data) could improve performance*
- *Some imbalance in feature distributions*

Conclusion:
*Data is **sufficient for reliable modelling**, but performance can improve with richer features.*

4.5 Linear vs Non-Linear Model Suitability

Findings

- *Linear models (e.g., Logistic Regression) showed limited performance*
- *Non-linear models (Decision Tree, SVM) performed better*

Reason

- *Relationships between features and target are **non-linear and complex***

Conclusion:
Non-linear models are more suitable for cardiovascular risk prediction due to their ability to capture complex interactions.

4.6 Final Model Selection Justification

The final model (SVM / tuned classifier) was selected based on:

1. High Recall (~80%)

- *Effectively identifies high-risk individuals*
- *Critical in healthcare to minimize missed diagnoses*

2. Strong Generalization

- *Train and test performance are consistent*
- *Reliable on unseen data*

3. Good ROC-AUC (~0.78)

- *Strong discrimination capability*

Trade-off

- *Lower precision (more false positives)*
- *Acceptable in this domain*

4.7 Final Outcomes of the Modelling Process

The modelling pipeline successfully achieved the business objective:

Model Performance Summary

- *Accuracy: ~70%*
- *Recall (High Risk): ~80%*
- *ROC-AUC: ~0.78*

Key Achievements

- *Reliable identification of high-risk individuals*
- *Balanced trade-off between sensitivity and specificity*
- *Robust and generalizable model*

4.8 Business Impact and Practical Implications

- *The model can be used as a **screening tool in healthcare systems***
- *Helps in:*
  - *Early detection of cardiovascular risk*
  - *Preventive healthcare planning*
- *Supports **data-driven medical decision-making***

4.9 Final Conclusion Statement

*The analysis demonstrates that cardiovascular risk prediction can be effectively addressed using machine learning techniques. By combining strong feature insights, appropriate model selection, and careful evaluation, the final model delivers a **reliable, interpretable, and practically useful solution**.*

*Most importantly, the model prioritizes **patient safety by maximizing recall**, making it well-suited for real-world healthcare applications where early detection is critical.*